

Polylogarithmic Approximation for Covering and Connecting Multi-Interface Networks

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Outline

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Motivation

- In IoT networks, each device may support multiple communication interfaces.
- Activating an interface consumes resources (energy, bandwidth, radio time).
- Goal: satisfy communication requirements while minimizing the maximum node cost.

Graph Model

- Input **graph** $G = (V, E)$ represents feasible physical links.
- Each **vertex** $v \in V$ has a set of available **interfaces**, $\lambda(v)$.
- For each interface $i \in \lambda(v)$, there is a **cost** $c(i, v)$ for activating it at v .
- We want to find an **assignment of active interfaces** $\lambda_A(v) \subseteq \lambda(v)$ for each v .
- There are some **communication requirements** that $\{\lambda_A(v)\}_{v \in V}$ must satisfy.
- Objective (min-max): **minimize**

$$\max_{v \in V} \sum_{i \in \lambda_A(v)} c(i, v),$$

Two Optimization Problems

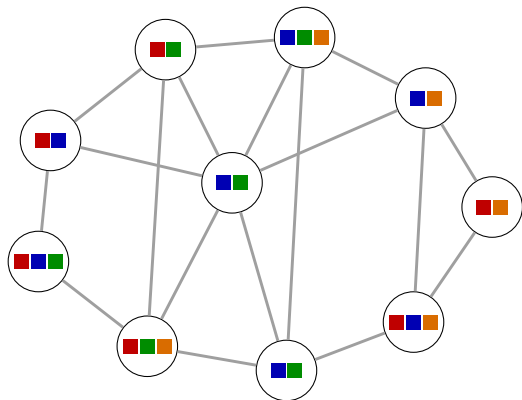
COVERAGE

Activate interfaces so every edge $uv \in E$ is covered by at least one active interface shared by u and v .

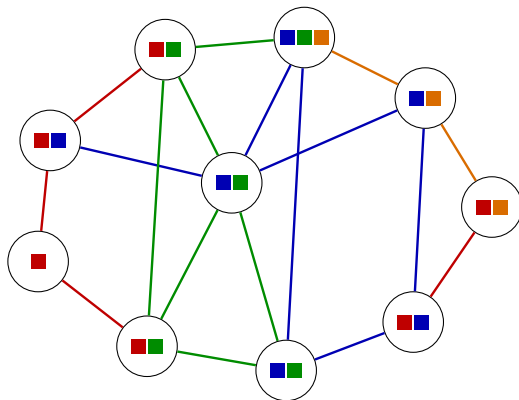
CONNECTIVITY

Given **terminal groups** $U_1, \dots, U_p \subseteq V$, activate interfaces so that for every $r \in [p]$, the subgraph induced by edges covered by active interfaces contains a path between every pair of vertices in U_r .

Illustration: COVERAGE

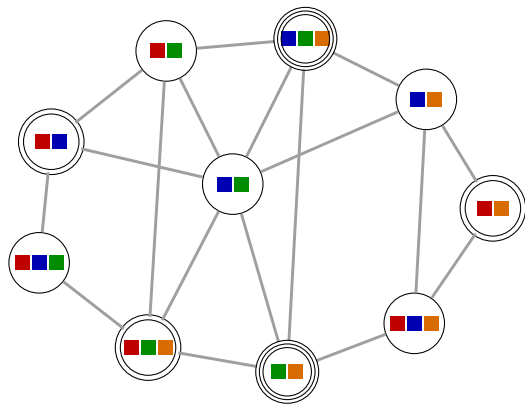


Colors represent available interfaces

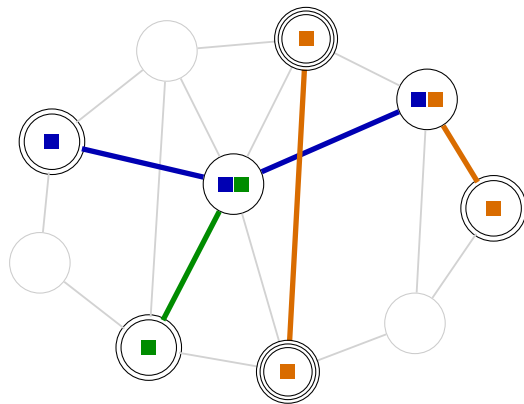


Colors represent active interfaces

Illustration: CONNECTIVITY

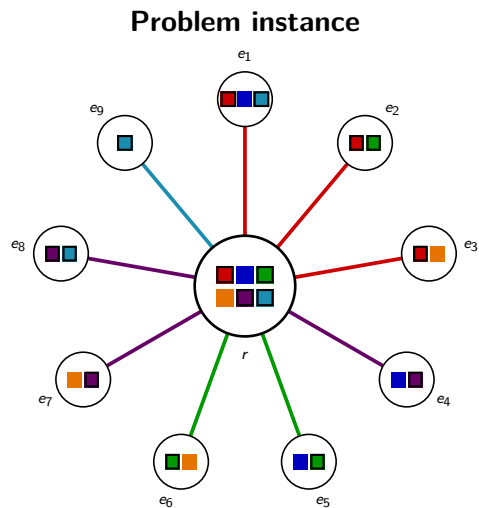
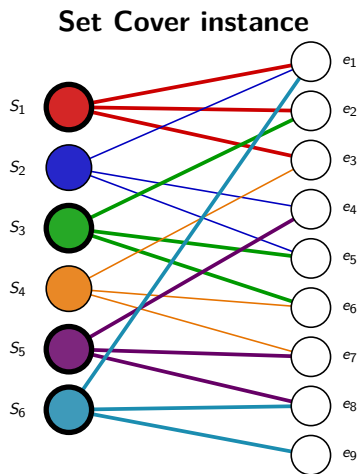


Contours represents terminal groups (double for groupe 1 and triple for group 2)



Colors represent active interfaces: only bold edges are covered

Inapproximability: Reduction from Set Cover



What is known in terms of approximation?

- Min-max COVERAGE, uniform costs: $O(b \cdot \log m)$ -approximation, where b is a certain graph parameter.
- Min-max CONNECTIVITY: no non-trivial approximation known.
- Min-sum COVERAGE has an $O(\sqrt{n} \log n)$ -approximation for the homogeneous cost case.
- Min-sum CONNECTIVITY has $3/2 + \epsilon$ -approximation for the homogeneous cost case and $O(\log n)$ -approximation for the heterogeneous cost case.

Preprocessing

- **Reason:** allows us to obtain lower bounds on OPT and enables concentration bounds to work.
- **Scaling of costs:** test consecutive powers of 2 and guess the smallest b such that the optimal solution uses interfaces of cost at most 2^b .
- Discard interfaces with cost larger than 2^b and divide all remaining costs by $\max_{v \in V(G), i \in \lambda(v)} \{c(i, v)\}$.
- After scaling: costs are in $[0, 1]$ and (for the right guess) $\text{OPT} \geq 1/2$.
- **Cheap vs expensive vertices:**
- If $\sum_{i \in \lambda(v)} c(i, v) \leq 1$, vertex v is cheap and we can activate all interfaces at v .
- Otherwise v is expensive, and we can assume that

$$\sum_{i \in \lambda(v)} c(i, v) \cdot x_{i,v} \geq 1.$$

Main Approximation Guarantees

- The COVERAGE problem with heterogeneous costs:

$O(\log m)$ -approximation

via LP relaxation + randomized rounding.

- Simpler deterministic rounding for the COVERAGE problem:

k -approximation

where k is the number of interfaces.

- The CONNECTIVITY problem with heterogeneous costs:

$O(\log^2 m)$ -approximation

via LP with flow/cut constraints + iterative randomized rounding.

Technique 1: ILP for the COVERAGE problem (paper formulation)

$$\begin{aligned}
 \min \quad & M \\
 \text{s.t.} \quad & \sum_{i \in \lambda(v)} c(i, v) x_{i,v} \leq M && \forall v \in V, \\
 & \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq 1 && \forall uv \in E, \\
 & \begin{cases} x_{i,v} = 1 \ \forall i \in \lambda(v) & \text{if } \sum_{i \in \lambda(v)} c(i, v) x_{i,v} < 1, \\ \sum_{i \in \lambda(v)} c(i, v) x_{i,v} \geq 1 & \text{otherwise} \end{cases} && \forall v \in V, \\
 & x_{i,v} \in \{0, 1\} && \forall v \in V, i \in \lambda(v).
 \end{aligned}$$

The function $\min\{x_{i,u}, x_{i,v}\}$ is non-linear, but can be linearized with additional constraints.

Technique 2: ILP for the CONNECTIVITY problem (paper formulation)

$$\min \quad M$$

$$\text{s.t.} \quad \sum_{i \in \lambda(v)} c(i, v) x_{i,v} \leq M \quad \forall v \in V,$$

$$\sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq y_{uv} \quad \forall uv \in E,$$

$$\begin{cases} x_{i,v} = 1 \quad \forall i \in \lambda(v) & \text{if } \sum_{i \in \lambda(v)} c(i, v) x_{i,v} < 1, \\ \sum_{i \in \lambda(v)} c(i, v) x_{i,v} \geq 1 & \text{otherwise} \end{cases} \quad \forall v \in V,$$

$$y(\delta(S)) \geq 1 \quad \forall r \in [p], \forall S \subset V : S \cap U_r \neq \emptyset, (V \setminus S) \cap U_r \neq \emptyset,$$

$$x_{i,v} \in \{0, 1\}, y_{uv} \in \{0, 1\}.$$

The LP relaxation can be solved by the ellipsoid method using a separation oracle for the cut constraints, implemented using max flow.

Pseudocode: Randomized Algorithm for the COVERAGE problem

Input: LP-relaxed solution $\{x_{i,v}\}_{i \in \lambda(v), v \in V}$.

Procedure

- ① Pick independent uniformly random thresholds $t_i \in [0, 1]$ for each $i \in [k]$.
- ② For every $v \in V$, set

$$\lambda_A(v) \leftarrow \{i \in \lambda(v) : 2 \ln m \cdot x_{i,v} \geq t_i\}.$$

- ③ Return $\{\lambda_A(v)\}_{v \in V}$.

Proof Outline for the COVERAGE problem: Main Claims

- ① **Feasibility:** For any edge uv , $\Pr[\lambda_A(u) \cap \lambda_A(v) = \emptyset] \leq 1/m^2$. By union bound over all m edges, Algorithm 1 returns a covering assignment with probability at least $1 - 1/m$.
- ② **Cost:** For each expensive vertex v , $\mathbb{E}[\text{cost}(v, \lambda_A)] \leq 2 \ln m \cdot \sum_{i \in \lambda(v)} x_{i,v} c(i, v)$. Chernoff with $\delta = 2$ gives

$$\Pr \left[\text{cost}(v, \lambda_A) \geq 6 \ln m \cdot \sum_{i \in \lambda(v)} x_{i,v} c(i, v) \right] \leq 1/m^2.$$

Union bound over vertices gives cost at most $6 \ln m \cdot \text{OPT}$ with high probability.

Pseudocode: Iterative Algorithm for the CONNECTIVITY problem

Input: LP-relaxed solution $\{x_{i,v}\}, \{y_{uv}\}$.

Procedure

- ① For every edge $e \in E$, set $p_e \leftarrow \min\{1, 4 \ln m \cdot y_e\}$
- ② Sample $Y_e \in \{0, 1\}$ with $\Pr[Y_e = 1] = p_e$.
- ③ Let H be the subgraph induced by edges with $Y_e = 1$.
- ④ Set $T \leftarrow \frac{2 \ln m}{1 - 1/e}$ and initialize $\lambda_A(v) \leftarrow \emptyset$ for all $v \in V$.
- ⑤ For each $j \in [T]$:
 - Pick independent uniformly random thresholds $t_i \in [0, 1]$ for each $i \in [k]$.
 - Set $\lambda_A^j(v) \leftarrow \{i \in [k] : 4 \ln m \cdot x_{i,v} \geq t_i\}$ for every $v \in V$.
 - Update $\lambda_A \leftarrow \lambda_A \cup \lambda_A^j$.
- ⑥ Return λ_A .

Proof Outline for the CONNECTIVITY problem: Main Claims

- ① **Terminal connectivity:** With high probability, for every terminal group U_r , graph H connects all terminals in U_r .
- ② **Progress per iteration:** If H_j is the uncovered part of H at iteration j , then $\mathbb{E}[|\text{cov}(\lambda_A^j, H_j)|] \geq (1 - 1/e) \cdot \mathbb{E}[|E(H_j)|]$.
- ③ **Number of iterations** Let I be the number of iterations required to cover all edges in H . Then $\Pr[I > T] = O(1/m)$.
- ④ **One-iteration cost:** For every fixed iteration j , $\text{cost}(\lambda_A^j) \leq 12 \ln m \cdot \text{OPT}$, with high probability.
- ⑤ **Total cost:** Union bound over all T iterations gives $\text{cost}(\lambda_A) \leq \frac{24 \ln^2 m}{1-1/e} \cdot \text{OPT}$.

Summary and Open Questions

- We gave $O(\log m)$ and $O(\log^2 m)$ approximations for the COVERAGE and CONNECTIVITY problems, respectively.
- For COVERAGE this is tight under standard complexity assumptions, but for CONNECTIVITY there is a gap between $O(\log^2 m)$ and the $\Omega(\log m)$ hardness from set cover.
- Can $O(\log^2 m)$ for the CONNECTIVITY problem be improved to $O(\log m)$?
- Conjecture: $O(\log^2 m)$ -approximation might be best possible, since the problem is similar to GROUP STEINER TREE which has $\Omega(\log^2 m)$ hardness.

Thank you!

Questions?